

Android Application Development

Assignment Question Sets — Set 1 & Set 2



Assignment Set 1 — Android Fundamentals & Architecture

Topics: *Mobile OS Overview · Android vs iOS · Android Features & Versions · Android Studio · Architecture · App Components · Manifest*

Unit 1 — Introduction to Mobile & Android

- Q1.** What is a Mobile Operating System? Compare Android and iOS across at least five parameters: openness, customization, app ecosystem, security model, and hardware compatibility.
- Q2.** Describe the major features of Android as a mobile platform. How have Android versions evolved from early releases to the latest — highlight at least four significant version milestones and the features they introduced.
- Q3.** What is Android Studio? Explain its key components (Project Structure, Layout Editor, Logcat, AVD Manager) and the steps involved in setting up Android Studio for the first time.
- Q4.** Explain how to run an Android application on both an Emulator (AVD) and a Physical Device. What is USB Debugging and why is it required for physical device testing?

Unit 2 — Android Architecture

- Q5.** Describe the five layers of Android Architecture: Linux Kernel, Libraries, Android Runtime, Application Framework, and Applications. What is the role of each layer?
- Q6.** What is the Android Runtime (ART)? How does it differ from the older Dalvik Virtual Machine? What advantages does ART offer in terms of performance?
- Q7.** Explain the Application Framework layer of Android Architecture. List at least five managers available in this layer and describe their responsibilities.

Unit 2 — Application Components & Manifest

- Q8.** What are the four core Application Components in Android? Give a one-line definition of each: Activity, Service, Broadcast Receiver, and Content Provider.
- Q9.** Explain the complete Activity Lifecycle in Android. Describe each callback method (onCreate, onStart, onResume, onPause, onStop, onDestroy) and when it is triggered.
- Q10.** What is the Android Manifest file (AndroidManifest.xml)? What information must be declared in it, and what happens if a component is not registered in the Manifest?
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Assignment Set 2 — UI Design, Storage & Modern Android

Topics: Layouts · Views & Widgets · Event Handling · Intents · Toasts/Dialogs/Menus · Data Storage · Kotlin · Cross-platform · APIs · Publishing

Unit 3 — UI Design & Layouts

- Q1.** Compare LinearLayout, RelativeLayout, ConstraintLayout, and GridLayout. For each, describe its positioning mechanism and give one practical use case where it is the best choice.
- Q2.** Describe the following Android widgets and their key XML attributes: TextView, EditText, Button, ImageView, and ScrollView. How are they declared in a layout XML file?
- Q3.** What is the difference between ListView and RecyclerView? Why is RecyclerView preferred for large datasets? What is the role of an Adapter in both?

Unit 3 — Event Handling & Intents

- Q4.** What is Event-driven Programming in Android? Explain the concept of Event Listeners and Callback Methods. Describe four event handling methods: onClick(), onKeyDown(), onTouch(), and onFocusChange().
- Q5.** What is the difference between an Explicit Intent and an Implicit Intent in Android? Explain how data is passed between two Activities using Intent extras (putExtra / getStringExtra).
- Q6.** Explain Toasts, AlertDialogs, and the three types of Menus in Android (Options Menu, Context Menu, and Popup Menu). How are each of these created and displayed?

Unit 4 — Data Storage

- Q7.** What are Shared Preferences in Android? How are they used to store and retrieve simple key-value data? What types of data can be stored using Shared Preferences?
- Q8.** Explain SQLite database in Android. What is the role of SQLiteOpenHelper? Describe all four CRUD operations (Insert, Read/Query, Update, Delete) with the relevant methods used.
- Q9.** What is Firebase Realtime Database? How do you set up Firebase in Android Studio and connect it to your app? Explain how to Insert, Retrieve, and Display data using Firebase.

Unit 5 — Modern Android & Publishing

- Q10.** What are the key features of Kotlin that make it preferred over Java for Android development? Briefly compare Flutter and React Native as cross-platform development frameworks. What is JSON Parsing and how is it used when consuming a REST API in Android?